

Chen Xu

Ph.D. Student, AIGC-DL Lab, University of Macau

[Email](#) | [Homepage](#) | [Google Scholar](#) | [GitHub](#)

Research Interests

- **Computer Vision**, with an emphasis on **Parameter-Efficient Fine-Tuning (PEFT)** of large models.
- **Diffusion and Flow-Matching Acceleration** for image generation, image restoration, and vision-language-action (VLA) models.

Academic Qualifications

Ph.D. Student, University of Macau

Macau

[AIGC-DL Lab](#) | Advisor: [Prof. Zhedong Zheng](#)

Bachelor of Engineering in Computer Science and Technology, Nankai University

[NKICS Lab](#) | Advisors: [Prof. Tao Li](#) and [Prof. Kai Wang](#)

Past Experience

Research Assistant, AIGC-DL Lab, University of Macau

Mar 2026 - Aug 2026

Institute of Machine Intelligence, Nankai University

Sep 2023 - Jun 2025

- Conducted research on TS-SAM, Resfusion, and MOST under Prof. Kai Wang and Prof. Tao Li; contributed to model design, training, evaluation, dataset development, and paper writing.

Undergraduate Innovation and Entrepreneurship Project, Nankai University

Apr 2023 - Apr 2024

- Led a competitive-sports video tracking project; generated pseudo-labels with SAM, curated a custom dataset, proposed MOST, and designed the lightweight MOST-SAM module.

AI Class Trainee, Chinasoft International Internship and Training

Jun 2023 - Jul 2023

- Led the Yellowbook game-forum project; built a local-server platform and trained a textRNN classifier with about 87% accuracy.

Research Paper Project, Yau Science Award Mathematics

Apr 2020 - Nov 2020

- Extended random interval graphs to higher-dimensional spaces and wrote "Research on Random Interval Graphs and Their Generalized Properties."

Awards

2024 **K Award (Participant)**, American Collegiate Mathematical Modeling Contest

2022 **Bronze Award**, Tianjin Volunteer Service Competition

2024 **Team Instructor**, U.S. Collegiate Mathematical Modeling Contest

2022 **Outstanding Team**, National Social Practice to the Countryside

2024 **Third Prize, Operating System Competition**, Computer System Development Capability Competition

2022 **Outstanding Social Practice Field Trip Report**

2023 **Fourth Place, Doubles**, 2023 Computer Science College Dean's Cup Badminton Tournament

2020 **First Prize, Northern Division; Finalist Award**, Yau Science Award Mathematics

2023 **Honorable Team**, Chinasoft International AI Class Internship and Training

2020 **Outstanding Student**, Shing-Tung Yau International Summer Math Camp for Secondary School Students

Publications

1. ScaleResfusion: Residual Rectified Flow based on Residual Vector Field.

arXiv preprint arXiv:2607.25275, 2026. [Paper](#) | [Code](#)

Zhenning Shi^{*}, **Chen Xu**^{*}, Junhao Zhang, Kefei Zhang, Linjie Liu, Zhedong Zheng, Tao Li.

2. Resfusion: Denoising Diffusion Probabilistic Models for Image Restoration Based on Prior Residual Noise.

Advances in Neural Information Processing Systems 37 (NeurIPS), 2024, pp. 130664-130693. [Paper](#)

Zhenning Shi, Haoshuai Zheng, **Chen Xu**, Changsheng Dong, Bin Pan, Xueshuo Xie, Along He, Tao Li, Huazhu Fu.

3. TS-SAM: Fine-Tuning Segment-Anything Model for Downstream Tasks.

2024 IEEE International Conference on Multimedia and Expo (ICME), 2024, pp. 1-6. [Paper](#)

Yang Yu, **Chen Xu**, Kai Wang.

^{*} Co-first authors (equal contribution).